

Non-Shock Initiation Model for Explosive Families: Experimental Results

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Abstract. The “DaMaGe-Initiated-Reaction” (DMGIR) computational model has been developed to predict the response of ideal high explosives to impulsive loading from non-shock mechanical insults. The distinguishing feature of this model is the introduction of a damage variable, which relates the evolution of damage to the initiation of a reaction in the explosive, and its growth to detonation. This model development effort treats the non-shock initiation behavior of explosives by families; rigid plastic bonded, cast, and moldable plastic explosives. Specifically designed experiments were used to study the initiation process of each explosive family with embedded shock sensors and optical diagnostics. The experimental portion of this model development began with a study of PBXN-5 to develop DMGIR model coefficients for the rigid plastic bonded family, followed by studies of the cast, and bulk-moldable explosive families, including the thermal effects on initiation for the cast explosive family. The experimental results show an initiation mechanism that is related to impulsive energy input and material damage, with well defined initiation thresholds for each explosive family. These initiation details will be used to extend the predictive capability of the DMGIR model from the rigid family into the cast and bulk-moldable families.

Introduction

The impulsive load caused by the impact of a projectile or fragment into an energetic material will produce a range of responses that can range from mechanical damage with no reaction to a self-sustaining deflagration that consumes the material at an increasing rate, transitioning into a detonation.

The magnitude of impulsive energy input, and the extent to which damage is generated in the energetic material have a direct effect on the onset

and the extent of reaction produced in an explosive sample. Impact-induced deflagrations with sufficient energy input and mechanical damage have been observed to transition within a few microseconds into a growth to detonation. The generation of damage is dependent on the shock loading and unloading profile, and the material strength. Numerical simulations of this damage are used to relate shear-induced strain to explosive initiation by the use of a damage variable which is embedded in the DMGIR non-shock initiation model. This treatment of non-shock initiation

allows all ideal explosives to be described numerically in one of the following three explosive families; rigid plastic bonded, cast, and moldable plastic explosives.¹

An experimental program was designed which made use of embedded shock sensors and optical diagnostics in order to better understand the explosive initiation process of these explosive families from impulsive, non-shock loading.

Experimental Description

A single-stage light gas, or a propellant gun was used in each test to accelerate a projectile to the desired velocity, impacting an explosive test specimen. Projectile free flight of ~8-feet allowed blast protection for the gun. The test specimen was either a solid or a sandwiched cylindrical billet, instrumented with PVDF piezoelectric shock sensors for velocity measurements.² Digital framing cameras recorded the details of projectile impact and subsequent reaction on each experiment. Thermal effects on the cast explosive family were evaluated at temperatures that ranged from ambient to 122C. The ambient temperature test configuration is shown schematically in Fig. 1.

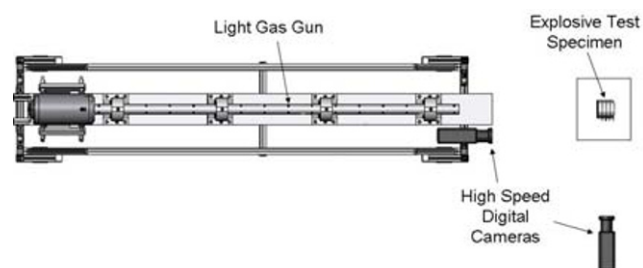


Fig. 1. Gun configuration for explosive study.

The projectile geometry consisted of a blunt-faced, hollow, right circular cylinder with a 0.500-inch diameter and an O-ring groove near the impact face. Projectile materials were aluminum or brass.

This projectile design provided a planar, short duration shock loading pulse, followed immediately by axial and radial rarefaction waves which provide well controlled mechanical damage, as shown in Fig. 2. This controlled damage was intended to simulate the expected profile produced by accidental impact.³ Impact velocity capabilities

range from 0.20 – 1.66 km/s. Explosively driven water blades were also used to generate mechanical damage, although less well controlled.

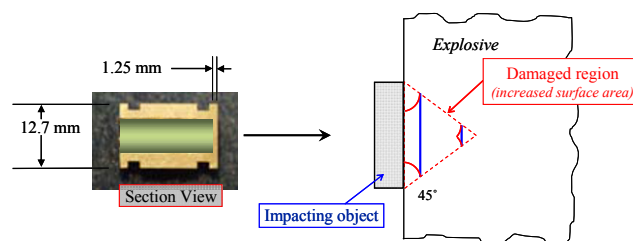


Fig. 2. Shock and release from projectile impact generates a controlled damage region in explosive.

Test Specimen

Ideal explosives are composite materials with explosive crystals embedded in a variety of polymer-based binders, with mechanical properties that depend mainly on the mixture used to form the polymer binder. The three explosive families described in this study were represented by: PBXN-5, $\rho=1.80$ gm/cc, (rigid plastic bonded), cast Composition-B, $\rho=1.61$ and 1.67 gm/cc, (cast), and PETN- or RDX-based Primasheet 1000, 2000, and Composition C-4, $\rho=1.60$ gm/cc (moldable plastic).

The ambient temperature test specimen configurations consisted of a cylindrical billet that was made from sandwiched billets with piezoelectric shock sensors embedded at each interface. Billet diameters were nominally 4.0–inch diameter. The piezoelectric shock sensors measured stress-time history along the axial centerline of the PBXN-5 billet. Shock front arrival times for each sensor location are plotted in Fig. 3 for a collection of PBXN-5 tests, showing the transition region between the mechanical response and the deflagration growth to detonation, as well as prompt detonation response from detonator input.